Solution:

$$\frac{2x-1}{4x^2} = \frac{1}{2x}+\frac{-1}{4x^2} = \frac{1}{4}\left[2\left(\frac{1}{x}\right)+\frac{-1}{x^2}\right]$$

Take $a = 2$,

$$f(x) = \frac{1}{x}$$

$$\therefore f'(x) = -\frac{1}{x^2}$$

$$\int e^{2x}\left[\frac{2x-1}{4x^2}\right]dx$$

$$= \frac{1}{4}\int e^{2x}\left[2\left(\frac{1}{x}\right)+\frac{-1}{x^2}\right]dx$$

$$= \frac{1}{4}\int e^{ax}\left[a\,f(x)+f'(x)\right]dx$$

$$= \frac{1}{4}\left[e^{ax}\ f(x)\right]+c$$

$$= \frac{1}{4x}\,e^{2x}+c$$

Example 2.39

Evaluate $\int\left[\frac{1}{\log x}-\frac{1}{(\log x)^2}\right]dx$

Solution:

Take $z = \log x$

$\therefore dz = \frac{1}{x}dx$

$\Rightarrow dx = e^z dz \quad \left[\because x = e^z\right]$

and $f(z) = \frac{1}{z}$

$\therefore f'(z) = -\frac{1}{z^2}$

$$\int\left[\frac{1}{\log x}-\frac{1}{(\log x)^2}\right]dx$$

$$= \int\left[\frac{1}{z}-\frac{1}{z^2}\right]e^z dz$$

$$= \int e^z\left[f(z)+f'(z)\right]dz$$

$$= e^z f(z)+c$$

$$= e^z\left[\frac{1}{z}\right]+c$$

$$= \frac{x}{\log x}+c$$

Exercise 2.6

Integrate the following with respect to x.

1. $\frac{2x+5}{x^2+5x-7}$
2. $\frac{e^{3\log x}}{x^4+1}$
3. $\frac{e^{2x}}{e^{2x}-2}$
4. $\frac{(\log x)^3}{x}$
5. $\frac{6x+7}{\sqrt{3x^2+7x-1}}$
6. $(4x+2)\sqrt{x^2+x+1}$
7. $x^8\left(1+x^9\right)^5$
8. $\frac{x^{e-1}+e^{x-1}}{x^e+e^x}$
9. $\frac{1}{x\log x}$
10. $\frac{x}{2x^4-3x^2-2}$
11. $e^x(1+x)\log\left(xe^x\right)$
12. $\frac{1}{x\left(x^2+1\right)}$
13. $e^x\left[\frac{1}{x^2}-\frac{2}{x^3}\right]$
14. $e^x\left[\frac{x-1}{(x+1)^3}\right]$
15. $e^{3x}\left[\frac{3x-1}{9x^2}\right]$

2.1.6 Some special types of Integrals

Type: VII

To evaluate the integrals of the form $\int\frac{dx}{ax^2+bx+c}$, $\int\frac{dx}{\sqrt{ax^2+bx+c}}$ and $\int\sqrt{ax^2+bx+c}\,dx$, first we have to express ax^2+bx+c as the sum or difference of two square terms [completing the squares], that is $(x+\alpha)^2+\beta^2$ (*or*) $(x+\alpha)^2-\beta^2$ (*or*) $\beta^2-(x+\alpha)^2$ and apply the suitable formula from the formulae given below.

1. $\int\frac{dx}{a^2-x^2} = \frac{1}{2a}\log\left|\frac{a+x}{a-x}\right|+c$
2. $\int\frac{dx}{x^2-a^2} = \frac{1}{2a}\log\left|\frac{x-a}{x+a}\right|+c$
3. $\int\frac{dx}{\sqrt{x^2-a^2}} = \log\left|x+\sqrt{x^2-a^2}\right|+c$
4. $\int\frac{dx}{\sqrt{x^2+a^2}} = \log\left|x+\sqrt{x^2+a^2}\right|+c$

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